

Permitting & Licensing Plan-Review Agent

FROM WEEKS AND MONTHS TO HOURS AND DAYS

An AI plan-review assistant that checks designs against local zoning and building codes before submission, flags issues for applicants to fix up front, and gives reviewers a structured first pass — so staff approve, not chase corrections. The issuance decision stays with a licensed human.

THE PROBLEM

- Reviewers manually check each plan against zoning and building codes.
- Incomplete submissions bounce back, adding correction cycles and weeks.
- Applicants wait without visibility; staff field status calls.
- Permitting delay shows up directly in the price of housing and development.

Weeks-months to issue rebuilding permits at the local pace California flagged as too slow

>26% share of a new single-family home's price driven by regulation (NAHB)

THE GOVERNED SOLUTION

- 1** Pre-check the plan
- 2** Flag for the applicant
- 3** Structure the review
-  **Human issues the permit** (*human gate*)
- 5** Record & audit

The agent accelerates the review; a licensed human issues the permit. Issuance is withheld in code.

THE ROI

Weeks → hours/days permit process with AI e-check, per the Governor of California

~50% zoning-review time saved with AI pre-check, City of Austin

25+ municipalities already using AI plan-review (Austin, Houston, Seattle, CO...)

Executive case: Every correction cycle removed is days off the clock for a resident or a builder — and permitting delay is a measurable housing cost. The agency keeps the same legal authority and reviewer judgment, but spends it on decisions instead of chasing incomplete paperwork.

SECURITY & COMPLIANCE

Identity & least privilege — Cognito + custom:slg_role; deny-by-default MCP gateway; effective scope = agent grant $\cap$ user entitlement.

Data protection — PII masking before inference; KMS encryption; S3 Object Lock (WORM) audit; private VPC + Bedrock VPC endpoint.

AI guardrails & HITL — Bedrock Guardrails; grounded, cited answers; consequential action withheld in code until a human approves.

Compliance posture — Bedrock FedRAMP High + DoD IL-4/5 in GovCloud; CJIS / IRS-1075 alignment; CloudTrail; ADA WCAG 2.1 AA.

Deploy: Standalone (own VPC/edge/identity/data/WORM audit) or + Whole-of-Government layer. Full CloudFormation + Terraform parity, commercial & GovCloud (US).